Assignment 1

Subject: Elements of Mechanical Engineering (ME2L501)

Last date for assignment submission: **14-02-2024**

1. At a particular point in a wooden member state of stress is as shown in Figure 1. The of the grain in the wood makes an angle of + 22° with the x axis. Determine the shear stress parallel to the grain.

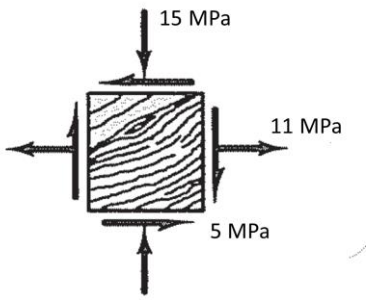

Figure 1

2. A rectangular beam is subjected to a pure bending moment M. The cross- section of the beam is shown in Figure 2. Determine the normal and shearing stresses at a point (x, y) on the plane AB shown.

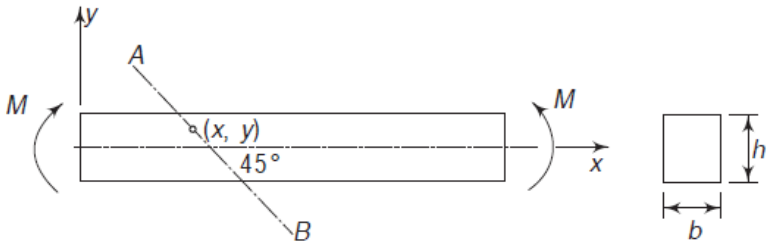

Figure 2

3. A rectangular bar of mild steel (as shown in the Figure 3) with a cross-sectional area of 4 cm × 3 cm experiences a tensile force of 7.6 kN. Calculate the normal and shear stresses on a plane whose normal is represented by the following direction cosines (l, m, n):

(a) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1$ (b) $0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ (c) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

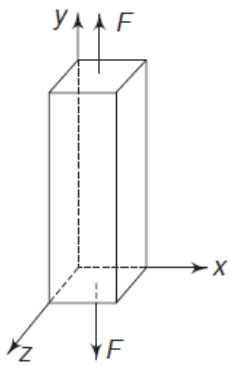

Figure 3

4. At a point P in a body, σ_x = 12 MPa, $\sigma_y = 8$ MPa, σ_z = –9 MPa, $\tau_{xy} = \tau_{yz} = \tau_{xz}$ = 6 MPa. Determine the normal and shearing stresses on a plane that is equally inclined to all the three axes.

5. The rectangular components of stress at a point are characterised by the following matrix:

$$[\sigma_{ij}] = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & -2 \\ 1 & -2 & -1 \end{bmatrix}$$

Determine the unit normal of a plane parallel to the x-axis (i.e. $l = 0$, on which the resultant stress vector is tangential to the plane).

6. The state of stress at a point is characterised by the components σ_x = 250 MPa, σ_y = -100 MPa, σ_z = –310 MPa, $\tau_{xy} = \tau_{yz} = \tau_{xz}$ = 0 Determine the extreme values of the shear stresses and their associated normal stresses.

7. Determine the principal stresses for the states of stress characterised by the following stress matrix:

$$[\sigma_{ij}] = \begin{bmatrix} 12 & -10 & 24 \\ -10 & -49 & 30 \\ 24 & 30 & 34 \end{bmatrix} \text{MPa}$$

8. Two stationary slabs made of rigid steel enclose a thin rubber sheet (see Figure 4). Friction between the rubber and steel faces is negligible. If the rubber plate is subjected to stresses σ_x = 24 MPa and $\sigma_y = 10$ MPa as shown, determine the strains ε_x and ε_y, and also the stress σ_z. E = 210 GPa and $\nu = 0.3$.

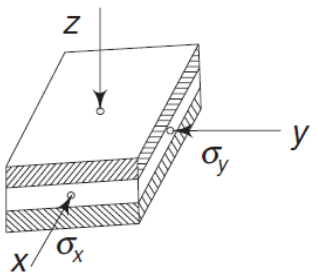

Figure 4

9. For the given strain matrix at a point, determine the stress matrix.

$$[\varepsilon_{ij}] = \begin{bmatrix} 0.0021 & 0 & -0.0017 \\ 0 & 0 & 0.0027 \\ -0.0017 & 0.0027 & -0.0039 \end{bmatrix}$$

Assume that the material is homogeneous and isotropic. E = 210 GPa and $\nu = 0.3$.

10. At a point in a stressed elastic plate, the following information is known: maximum shear strain = 390 µm/m, and the sum of the normal stresses on two perpendicular planes passing through the point is 50 MPa. The elastic properties of the plate are E = 205 GPa and G = 82 GPa. Calculate the magnitude of the principal stresses at the point.